

Plume-Surface Interaction: Physics Notes

1 mm nozzle plume impinging on a flat surface — interpretation of density, translational temperature, velocity, surface pressure, surface shear and heat flux

Companion notes for the AMR DSMC simulation results (den112, trans112, velocity112) in openfoam_mach3_cylinder / comet_result / output_amr3_result / field.

1. Slide-ready talking points (3 headline bullets)

Slide 1 — Plume expansion and recompression shock

Files referenced: den112.png, trans112.png.

- Centerline density drops by orders of magnitude from nozzle (Location 3) to plate (Location 1) — rapid near-field expansion of the 1 mm plume.
- Deep cold spot on the axis followed by a broad T_x peak at Locations 3/4/2 — barrel/reflected shock recompression, not free expansion alone.

Slide 2 — Impingement builds a wall-jet on the plate

Files referenced: velocity112.png.

- At the surface (Location 1): $U \approx 0$ on axis (stagnation), rising to an off-axis maximum — classic wall-jet after impact.
- Upper probes (Locations 3/4/2) still show a jet-core peak on axis — the momentum source that drives the surface loading below.

Slide 3 — 2D axisymmetric $\equiv$ 3D for this geometry

Files referenced: all three plots.

- Solid (2D) and dashed (3D) curves nearly overlap at every height in ρ , T_x , and U .
- 2D is sufficient for parametric PSI sweeps; 3D only required if azimuthal asymmetry (tilted plate, off-axis nozzle) is introduced.

2. The master picture (keep this in mind)

A 1 mm nozzle fires hot dense gas into a near-vacuum. The gas:

- **Expands** at the lip (gets fast and cold).
- **Over-expands** and is pushed back by a **barrel shock** (gets hot and slower).
- **Impinges** on a flat plate where it stops axially and turns radially (stagnation + wall jet).
- **Delivers** pressure, friction, and heat to the plate.

Every curve you plotted is one snapshot of that story.

3. Why each profile looks the way it does

3.1 Density (ρ)

What it is: mass per unit volume of gas.

What drives its shape:

- At the nozzle exit, ρ is highest — the gas is still confined.
- As the plume moves downstream, it spreads radially into a larger area, so mass per volume falls.
- The expansion wave also reduces density (gas spreading into vacuum).
- Orders-of-magnitude drop from nozzle to plate is expected — conservation of mass working against geometry.
- At large radius, all curves collapse to ambient because the plume no longer dominates.

Q&A line: "Density drops because the plume is expanding into a vacuum; the 1 mm core spreads into a much larger radial area, and mass conservation forces ρ to fall."

3.2 Translational Temperature (T_x)

What it is: the temperature derived from the directed random motion of molecules in the x-direction. In DSMC it is reported separately from T_y , T_z because they can differ out of equilibrium.

What drives its shape:

- **Cold dip on the axis:** the expansion fan converts thermal energy into directed kinetic energy — thermal motion becomes directed motion, temperature drops.
- **Temperature peak after the dip:** the over-expanded plume is recompressed across the barrel shock; directed KE is converted back into random thermal energy, so T jumps.
- **Relaxation to ambient at large radius:** the plume gas mixes with quiescent ambient.
- **At the plate (Location 1)** the curve is nearly flat near ambient because the wall-jet has mixed and cooled the flow.

Q&A line: "The cold dip is expansion cooling (thermal $\rightarrow$ kinetic), the peak is the barrel-shock recompression (kinetic $\rightarrow$ thermal). Two halves of an under-expanded plume."

3.3 Velocity (U)

What it is: the bulk flow speed of the gas.

What drives its shape:

- **High core velocity at Locations 3/4/2:** the expansion fan accelerates the gas — cooling and acceleration are coupled.
- **Velocity drop after the peak:** the barrel shock slows the flow (compressions always decelerate supersonic flow).
- **Decay outward from the axis:** the plume has less momentum and more mixing with ambient as r grows.
- **At the plate, $U \approx 0$ on the axis:** the stagnation point; axial momentum is completely stopped by the plate.
- **Off-axis on the plate, U rises again:** this is the wall-jet, the axial flow turned 90° to slide along the surface.

Q&A line: "Upper probes show the accelerated jet core from expansion; the plate profile shows a stagnation point on the axis and a wall-jet off-axis — axial momentum is rotated 90° when it hits the surface."

4. Surface loading on the plate

4.1 Surface pressure p_{wall}

What it is: normal force per unit area on the plate.

- **Maximum at the stagnation point ($r=0$):** axial momentum is fully destroyed by the plate — that kinetic energy becomes stagnation pressure on the surface.
- **Monotonic decay outward:** the radial wall-jet slides along the surface and exerts almost no normal force.
- The shape is roughly a bell curve centered on the axis, with width comparable to the plume diameter at the plate.

Q&A line: " p_{wall} peaks at the stagnation point because that is where all the axial momentum is destroyed by the surface, and decays outward because the wall-jet is parallel to the surface and exerts no normal force."

4.2 Surface shear stress τ_{wall}

What it is: tangential (friction) force per unit area on the plate — the force the gas drags along the surface.

- **Zero at the stagnation point ($r=0$):** by symmetry, there is no tangential velocity at the axis. No flow along the surface $\Rightarrow$ no friction.
- **Rises to a peak off-axis:** where the wall-jet is fully developed, the near-wall velocity gradient du/dy is maximum, and so is viscous shear.
- **Decays outward:** the wall-jet slows, its boundary layer thickens, du/dy falls.
- The shape is the characteristic "zero—peak—decay" curve centered on the stagnation line.

Q&A line: " τ_{wall} is zero on the axis because there is no tangential flow at the stagnation point, peaks where the wall-jet is fastest, and decays as the wall-jet spreads and loses momentum."

4.3 Heat flux q_{wall}

What it is: thermal energy per unit area per unit time transferred from the gas to the plate.

Why q_{wall} peaks at the stagnation point:

- Stagnation gas is at the highest temperature (full kinetic $\rightarrow$ thermal conversion at the impact point).
- The thermal boundary layer is thinnest there, so dT/dy is largest.
- Density is also highest — more molecules hit the wall per second.

Decays outward: the wall-jet cools, boundary layer thickens, density falls — all three reduce dT/dy and therefore q_{wall} .

In DSMC you also multiply by the thermal accommodation coefficient α : molecules do not always fully exchange energy with the wall in a single collision, which is why rarefied q_{wall} peaks are lower than continuum predictions.

Q&A line: " q_{wall} peaks at the stagnation point where the gas is hottest, the boundary layer is thinnest, and density is highest; it decays outward for the same reasons. For rarefied flow the peak is scaled by the thermal accommodation coefficient."

5. Expansion wave vs barrel shock

They are **two different things**, and **both are in your plume**. They just occur at different stages of the same flow.

Aspect	Expansion wave (Prandtl–Meyer fan)	Barrel shock
Nature	Smooth, isentropic (reversible)	Shock — discontinuous, non-isentropic (irreversible)
Effect on gas	Accelerates, cools, density drops	Decelerates/redirects, heats, density rises
Location	At the nozzle lip and through the near-field core	Curved shock wrapping the plume boundary, further downstream
Signature in your plots	Cold spot / density drop on the axis at Locs 3/4/2	T_x peak after the cold spot, density recovery, U drop

The sequence in an under-expanded nozzle plume

- **Nozzle lip:** $p_{\text{nozzle}} \gg p_{\text{ambient}}$. Prandtl–Meyer fan fires radially outward; core gas turns, accelerates, cools.
- **Jet boundary:** the expansion fan reflects off the constant-pressure contact surface and returns as compression.
- **Compression coalesces into a shock** → the barrel shock. Gas crossing it is compressed, heated, slowed.
- **Mach disk (optional):** if pressure ratio is high enough, the barrel shock terminates in a normal shock on the axis.
- Post-shock gas then impinges on the plate.

“The expansion fan and the barrel shock are not the same — they are the two halves of the same story. The expansion fan cools and accelerates the core (cold spot), and the barrel shock recompresses it (temperature peak). Both have to be present in an under-expanded plume — if only one were, mass and momentum would not balance.”

6. How do we know it really is a barrel shock?

Three pieces of evidence, all of which must be true for a barrel shock. This simulation satisfies all three.

Evidence 1 — Geometry forces under-expansion

A 1 mm nozzle firing into a near-vacuum ambient is automatically under-expanded. That guarantees an expansion fan at the lip and a recompression shock downstream — it is not a guess, it is a conservation requirement.

Evidence 2 — The T_x profile has the canonical dip-then-peak signature

In `trans112.png`, Locations 3/4/2 show:

- A cold minimum near the axis $\rightarrow$ expansion fan (isentropic cooling).
- A broad hot peak off-axis $\rightarrow$ gas crossing a recompression shock.

A free, unshocked plume would show only the cold core monotonically relaxing to ambient. The presence of a peak is the fingerprint of a shock wave, and the fact that the peak lies **off the axis** tells you it is an oblique/curved shock wall, not a normal shock on axis.

Evidence 3 — The peak moves radially with axial distance

The radial position of the T_x peak at Locations 3, 4, 2 is different at different heights. A planar shock would show the peak at the same radius at every height. A peak that migrates radially with height is a **curved shock** — exactly what a barrel shock is.

Cross-check from the other two plots

- **den112**: density collapses through the expansion fan and shows a small local recovery at the same radial position where T_x peaks $\Rightarrow$ consistent with a compressive shock.
- **velocity112**: velocity drops sharply at the same radial location $\Rightarrow$ consistent with a shock, which must decelerate supersonic flow.

Ruling out the alternatives

Alternative	Would look like	Why ruled out
No shock (pure expansion)	Monotonic cold core relaxing to ambient	We see a temperature peak
Mach disk (normal shock on axis)	T peak on the axis ($r=0$)	Our peaks are off the axis
Bow shock from the plate	Peak just above the plate at Location 1 only	Our peaks are at Locs 3/4/2 upstream; Loc 1 is nearly flat
Contact-surface mixing only	Gradual blending, no sharp peak	We have a sharp peak, not gradual mixing

“A barrel shock is the curved oblique recompression shock that wraps around the core of an under-expanded plume. I’m sure mine is a barrel shock because (a) a 1 mm nozzle into vacuum is automatically under-expanded so a recompression shock is required, (b) the T_x , density and velocity profiles show a sharp off-axis jump — the signature of a shock, not of free

mixing, and (c) the radial position of that jump moves with axial height, which means the shock wall is curved. Planar result would be a Mach disk; off-axis, height-dependent result is a barrel shock."

7. What a wall jet is, and how it develops

Definition: a wall jet is the thin, fast-moving layer of gas that flows parallel to a surface after an impinging jet has been turned by that surface.

“A wall jet is the radial stream of fluid that forms after an axial jet impinges on a surface, because the jet's axial momentum cannot pass through the wall and is redirected sideways along it.”

Development — the four stages

- **Stage 1 — Free jet approach:** the jet is still moving axially, nothing special at the wall yet.
- **Stage 2 — Stagnation region ($r \approx 0$):** the plate blocks axial motion; flow decelerates to zero on the axis. Axial momentum becomes high surface pressure and a radial pressure gradient.
- **Stage 3 — Turning region (within ~ 1 jet diameter):** the radial pressure gradient accelerates the fluid outward; the flow is turned 90° from axial to radial. By the end of the turning region, the jet is a thin sheet flowing radially along the plate — the wall jet.
- **Stage 4 — Wall-jet region (beyond ~ 1 jet diameter):** the jet has its characteristic two-layer structure.

Two stacked layers of the wall-jet

- **Inner layer (wall $\rightarrow$ velocity peak):** a classical viscous boundary layer; velocity zero at the wall, rises to a maximum a short distance above. Shear stress and heat flux are high here because the gradients of u and T are steepest.
- **Outer layer (velocity peak $\rightarrow$ ambient):** a free shear layer that mixes with and entrains the ambient gas above.

Why it exists (two conservation laws)

- **Mass conservation:** mass arrives continuously at the stagnation region, cannot accumulate (steady state) and cannot go through the plate; the only escape is along the plate.
- **Momentum conservation:** incoming axial momentum has to go somewhere. Part becomes normal force on the plate, the rest is rotated into the radial direction and carried away by the wall jet.

Why it matters for PSI

Quantity	Peaks where	Why
p_{wall}	Stagnation point ($r=0$)	Full axial momentum destroyed there
τ_{wall}	Off-axis, inside the wall-jet region	du/dy is steepest where the wall-jet is fully established
q_{wall}	Stagnation point ($r=0$)	Thinnest thermal boundary layer + hottest stagnation gas

“A wall jet is the thin radial layer that forms along the plate after the plume impinges, because mass and momentum have no other escape route than to turn and flow along the surface. It has an inner viscous boundary layer plus an outer free-shear layer, and it is what sets the distribution of shear and heat flux on the plate away from the stagnation point.”

8. The single sentence that ties everything together

“Density falls because the plume expands into vacuum. Temperature first falls (expansion cooling) then rises (barrel shock recompression) then relaxes to ambient. Velocity accelerates through the expansion fan, slows across the shock, and goes to zero on the plate axis where it turns into a radial wall-jet. That stagnation point is where the plate gets the most pressure and the most heat; the wall-jet off-axis is where it gets the most shear. Every curve on every plot is one of those four stages — expansion → shock → stagnation → wall-jet.”

Q&A escape hatch

If someone asks “why does your curve look like X?” and you are not sure, fall back to:

“That follows from the chain: expansion → shock recompression → stagnation → wall-jet. Whichever of those four stages dominates at that probe location sets the local $\rho/T/U$, and whichever of the stagnation vs. wall-jet region dominates at that radius sets the local $\rho_{wall}/\tau_{wall}/q_{wall}$ ”

That answer is physically correct for every point on every one of your curves, so you cannot be cornered.